COP 3835 / FINAL PROJECT

Performance & Testing Report

Kailer Fernandez Nadal | Published portfolio: https://kailerf.github.io/ | November 11, 2026

1. Lighthouse audit - published website

Lighthouse 13.6.0, headless Chrome 158. Two real published-URL runs were recorded on November 11, 2026 (mobile: 18:52:44 UTC; desktop: 18:52:56 UTC). The audits reported no run warnings.

Category	Mobile	Desktop
Performance	99 / 100	100 / 100
Accessibility	100 / 100	100 / 100
Best Practices	96 / 100	96 / 100
SEO	100 / 100	100 / 100

These are measured Lighthouse scores for the public URL, not estimates or local-only scores. Mobile uses Lighthouse mobile emulation; desktop uses its desktop preset.

2. Recorded loading metrics

Metric	Mobile	Desktop
First Contentful Paint	0.9 s	0.2 s
Largest Contentful Paint	0.9 s	0.3 s
Speed Index	0.9 s	0.2 s
Total Blocking Time	90 ms	0 ms
Cumulative Layout Shift	0	0

The website displayed low layout shift (CLS 0) in both audits. On mobile, Lighthouse reported approximately 0.9 s for FCP/LCP and 90 ms for Total Blocking Time. Your browser and connection may yield different timings.

3. Identified issue and concrete optimization performed

Issue: The original portrait was a relatively large PNG for a web portfolio. **Optimization:** The image was converted and compressed to JPEG while keeping its 575 x 577 dimensions. The public Lighthouse audit retrieved the optimized image.

Image version	Measured size	Meaning
Original uploaded picture .png	387,768 bytes	Original source file
Deployed images/profile.jpg	35,642 bytes	Optimized public website JPG
Reduction	90.8%	352,126 bytes smaller

Project images also use lazy loading. This optimization preceded the submitted Lighthouse measurements; no before-and-after Lighthouse score comparison is claimed.

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Published-Site Verification

Two browsers / three viewport widths / evidence-based review

4. Published website: Chromium and Firefox

A GitHub Actions Playwright workflow loaded **https://kailerf.github.io/** in **Chromium** and **Firefox**. Each browser tested a 900-pixel-high viewport at widths of 375, 768, and 1280 pixels. The user-provided result files and JPG screenshots document each of the six browser/viewport combinations.

Viewport	Chromium	Firefox	Overall
375 px - phone	11 / 11	11 / 11	Passed
768 px - tablet	11 / 11	11 / 11	Passed
1280 px - desktop	10 / 10	10 / 10	Passed
TOTAL	32 / 32	32 / 32	64 / 64 passed

Result: 64 of 64 checks passed. Neither browser reported uncaught JavaScript errors or page-wide horizontal overflow at any of the tested widths. Screenshot files were reviewed and show responsive sections, portrait, project cards, skills table, and the contact form.

5. Browser verification: what was checked

Both browsers returned HTTP 200, displayed all five required sections and three JavaScript-generated projects, decoded website images, and fitted the skills table without clipping. On phone and tablet widths, opening the navigation and closing it with Escape passed; on desktop, the horizontal navigation was visible. Theme switching and invalid contact-form feedback also passed.

Testing boundary: The automated form test intentionally submitted empty values to the **Test validation only** control; it did not send email or validate FormSubmit inbox delivery. The JPG evidence shows this expected error-message state. The browser tests did not separately evaluate every possible keyboard or assistive-technology interaction.

6. Remaining Lighthouse findings (not failed browser tests)

favicon.ico request (404): Both published Lighthouse reports identified a missing website favicon. Lighthouse reported **Best Practices 96/100**; the favicon error was logged in the console. A JPG favicon link in the HTML head would avoid the implicit .ico request while respecting the JPG-only image rule. Rerun Lighthouse after changing the site.

Accessible name differences: Lighthouse flagged a mismatch between visible text and accessible names for the header brand link ("KF.") and theme button ("Light mode"). The overall Accessibility score still displayed 100; addressing these mismatches would improve voice-control usability.

7. Testing conclusion and evidence

Completed: published Lighthouse mobile and desktop audits; two-browser Chromium/Firefox published-site checks at 375, 768, and 1280 px; documented image optimization. The original separate local verification also passed 90 / 90 checks, but the stronger live-site evidence above is used for the browser requirement.

Files reviewed: lighthouse-mobile.report.json; lighthouse-desktop.report.json; corresponding Lighthouse HTML files; chromium-results.json and firefox-results.json; chromium-375/768/1280.jpg and firefox-375/768/1280.jpg; earlier image-size measurements. Browser engine build numbers were not recorded by the supplied two-browser JSON logs.